Spectroscopy and Atomic Physics

01.04 · Original practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

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A MAIS Physics & Astronomy Club study resource. These are original exercises, not official IOAA problems.

PROBLEMS

01 Reading a Doppler shift

A laboratory spectral line at 500.0 nm is observed at 500.2 nm in a distant star. Assume $|v| \ll c$ and use $\Delta\lambda/\lambda \approx v/c$ with $c = 3.00 \times 10^8$ m s^{-1}. (a) Find the line-of-sight speed and state whether the star is approaching or receding. (b) Can this measurement alone determine the star's full space velocity? Explain.

02 Energy levels and lines

An atom has energy levels -13.6 eV, -3.4 eV and -1.51 eV. Use $hc = 1240$ eV nm. (a) Find the wavelength of the photon emitted in a transition from -3.4 eV to -13.6 eV. (b) Find the wavelength for the transition from -1.51 eV to -3.4 eV. (c) Explain why the same transitions can appear as absorption lines under suitable conditions.

03 Line formation and broadening

A hot, dense source is viewed through a cooler, dilute gas containing an element that absorbs at a particular wavelength. (a) Describe the spectrum seen against the continuum. (b) Give two distinct mechanisms that can broaden a spectral line and describe how each works. (c) Why might the line be shifted as well as broadened?

04 Scattering and polarization

Unpolarized light of intensity I_0 passes through an ideal linear polarizer, then through a second ideal polarizer whose axis is at 60° to the first. Use Malus's law $I = I_{in}\cos^2\theta$ for already linearly polarized light. (a) Find the final intensity in terms of I_0. (b) Explain qualitatively how scattering can polarize starlight. (c) What happens to the final intensity if the polarizer axes are perpendicular?